

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1167

Battlefield 138 kV
April 17, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1167. The Interconnection Customer proposes a 72.7 MW solar photovoltaic facility interconnecting at the Battlefield 138 kV substation in the Virginia area. The request was received on March 12, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 438 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,766,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Toano – Lightfoot 138 KV	Yarnen – Mira Vista 345 KV line	140.9	335	53.2
Lightfoot Substation – Synata Hugoton 138 KV L	Yarnen – Mira Vista 345 KV	135.2	376	53.2
Battlefield – Synata Hugoton 138 KV	Mira Vista – Yarnen 345 KV L	131.5	409	53.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Toano – Lightfoot 138 KV Line	Rebuild line with higher-capacity conductor	\$1,929,000	15% (\$297,000)
Lightfoot Substation – Synata Hugoton 138 KV L	Rebuild line with higher-capacity conductor	\$8,012,000	15% (\$1,232,000)
Battlefield – Synata Hugoton 138 KV	Rebuild line with higher-capacity conductor	\$1,541,000	15% (\$237,000)
	Total allocated to project		\$1,766,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.